

Uncovering and Managing the Precursors That Negatively Impact Safety, Quality, and Productivity



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Overview

- Significant Safety and Quality Risks
- Introduction to Precursors
- Precursors Converge to Create Risk
- Don't Skip the Deep Dive
- Precursors Can Be Identified Pre-Incident

Theme: Adapt the Tools and Terminology to YOUR Needs



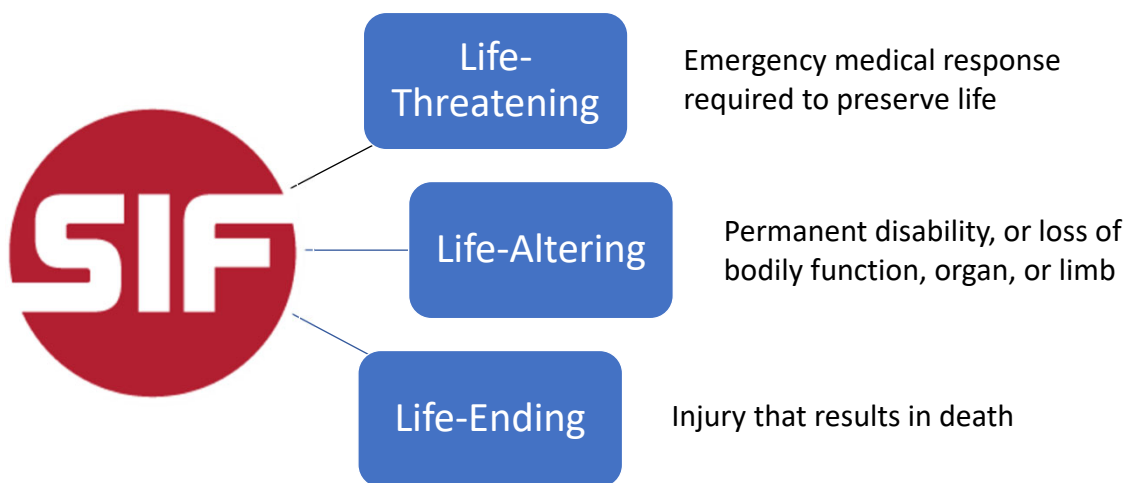
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Significant Safety & Quality Risks



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Serious Injuries and Fatalities (SIF)



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Life-Saving Commitments

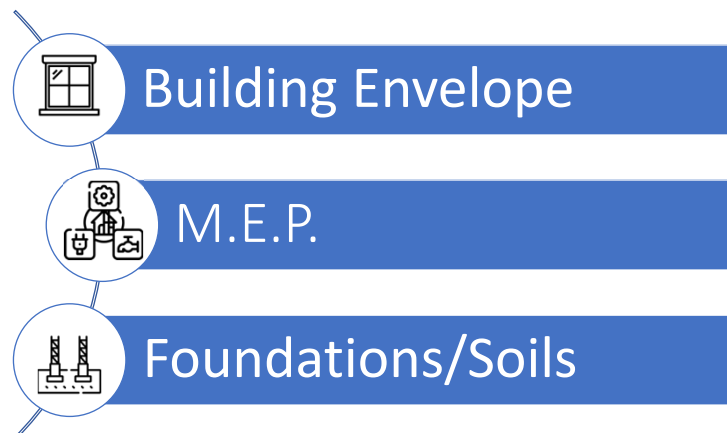
Where SIF's most frequently occur, within ACIG

↑ Energy = ↑ Hazard



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Top 3 Construction Defect Claims by %



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Introduction to Precursors



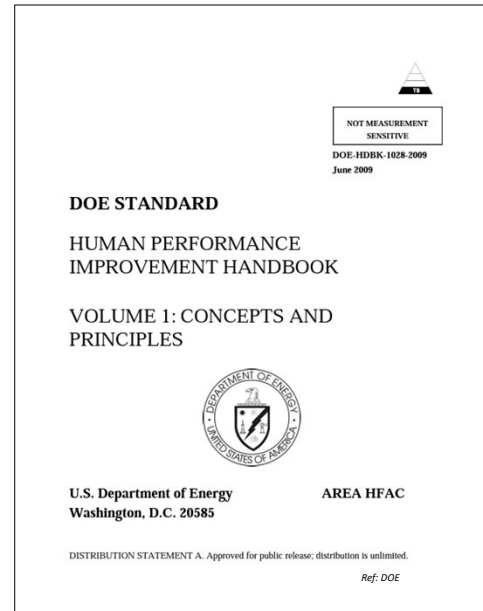
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Why Do Employees or Crews Have Incidents / Injuries?



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“Events can be avoided through an understanding of the reasons mistakes occur”



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Why Do Employees or Crews Have Installation Defects?



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Terminology

- Precursors are factors that contribute to a higher probability for error and SIF exposure
- Precursors are *identifiable* at the crew-level *before* a SIF occurs.

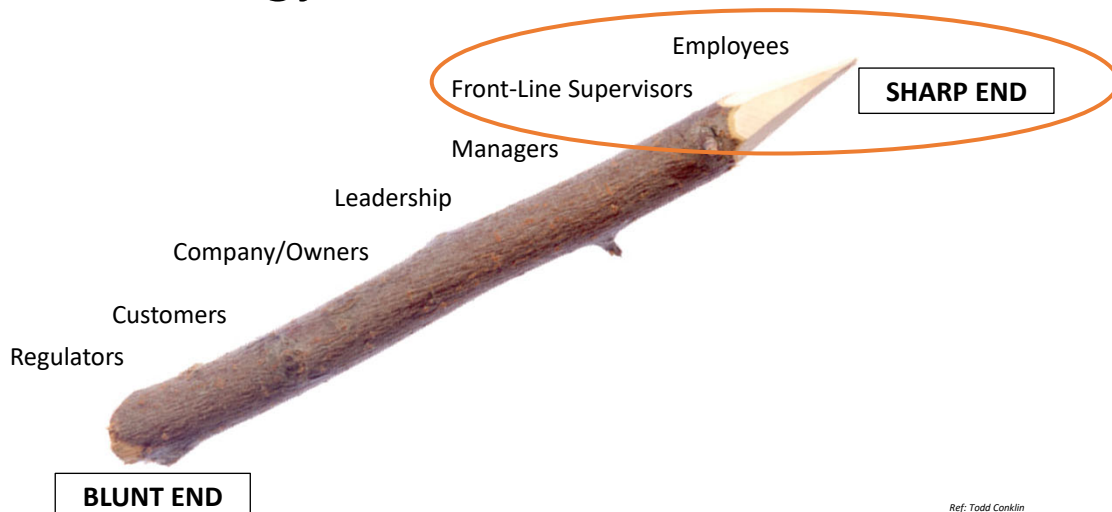
AKA:

- Risk multipliers
- Risk factors
- Red flags
- Error traps
- Stressors
- Constraints
- Challenges



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Terminology



Ref: Todd Conklin



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Terminology



Ref: Jaynes Corp.

Where "Precursors"
are found



Ref: Valiant Group

Where "Root Causes"
are (usually) found



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Financial Incentives
Staffing Levels
Commitment / Culture
Equipment / Tool Investments
Safety and Training Budgets
Risky Ventures



In a Hurry
Fatigue / Wellness
Supervision Over Capacity
Procedures not Implemented
Lack of Understanding
Lack of Planning



Root Causes

Influenced by Leadership

Precursors

Inherited by Employees



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Where SIF's Occur



Why SIF's Occur



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CHANGE MANAGEMENT	INDIVIDUAL FACTORS	PLANNING/ PREPAREDNESS	PROCESS/ PROCEDURE	SCHEDULING/ CAPACITY	TRAINING/ EXPERIENCE	WORK ENVIRONMENT	CONTROLS
What changed? What was atypical?	What was the employee dealing with?	Were they fully prepared?	Were clear guidelines provided for the work?	Were they trying to save time? Spread too thin?	Was training and tenure ideal?	What was going on around them?	What control break downs contributed?
1. Rework 2. Breakdown or Emergency Maintenance 3. Change in Personnel (Crew or Supervision) 4. Change in Role or Responsibility 5. Change in Task or Method 6. Change in Scope, Design, or Materials 7. Interruption or Unplanned Event Other:	8. Distraction or Mind Not on Task 9. Fit for Duty – Drugs or Alcohol 10. Fit for Duty – Physical or Mental Wellness 11. Discomfort With Intervention / Stop Work Other:	12. Communication Breakdown – Language Barrier 13. Communication Breakdown – Unclear Work Instructions 14. Pre-Planning – Not Completed, or Not Completed in Work Area 15. Pre-Planning – SIF-Risk / Quality Exposures Not Addressed 16. Pre-Planning – Work Changes Not Addressed 17. Tools / Equipment – Availability, Selection, or Condition 18. Unfamiliar or Non-Routine Activity Other:	19. Lack of Authorization or Permit-Required Work Failure 20. Process / Procedure – Not Implemented or Available in Field 21. Process / Procedure – Inadequate, Unclear, or Not Workable 22. Repetitive Work; Monotony Other:	23. Improper Work Sequencing or Trade Stacking 24. In a Hurry / Production Pressure 25. Lack of Oversight for Significant Activity 26. Non-Conventional Work Shift 27. Significant Overtime (> Typical Working Hours) 28. Supervision – Over Capacity or Not Present 29. Working Alone or Inadequate Crew Size for Activity Other:	30. Crew Member – Lack of Time in Job or On-Site (Tenure) 31. Crew Member – Lack of Training or Understanding for Task at Hand 32. Life-Saving Commitments / Key Rules Not Understood 33. Supervision – Lack of Training or Understanding 34. Supervision – Tenure (Less Than One Year in Position) 35. Inspector or Mgr. – Lack of Training or Experience Other:	36. Adverse Weather – High Winds, Lightning, Ice, Fog, etc. 37. Ambient Factors – Temperature, Humidity, Lighting, Noise, Air Quality 38. Confusing Displays and Controls 39. Work Area – New or Not Evaluated / Inspected 40. Work Area – Congested Workspace / Crowding 41. Limited Access for Inspection Other:	42. Limited Margin for Error 43. Line of Fire Uncontrolled 44. Lack of Visual Warning and / or Control Barrier 45. Motor Vehicle / Equipment – Condition and Maintenance List Other Hazard or QA/QC Controls Missing or Eroded: 46. 47.



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SIF-Risk Activities and Precursors Don't Mix!



= LSC



Ref: Tenor



= Precursors



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Precursors Converge to Create Risk



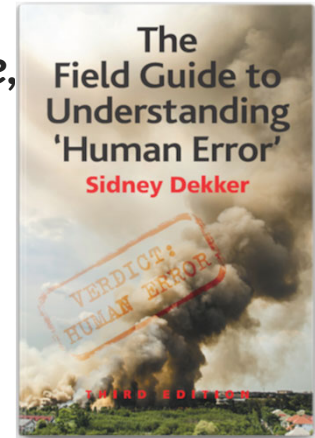
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It's Never One Thing

“There is no single cause - neither for failure, nor for success. In order to push a well-defended system over the edge... a large number of contributory factors are necessary and only jointly sufficient...”

A lot needs to go wrong for an incident or accident to occur.”

- Dr. Sidney Dekker



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Jenga® Model

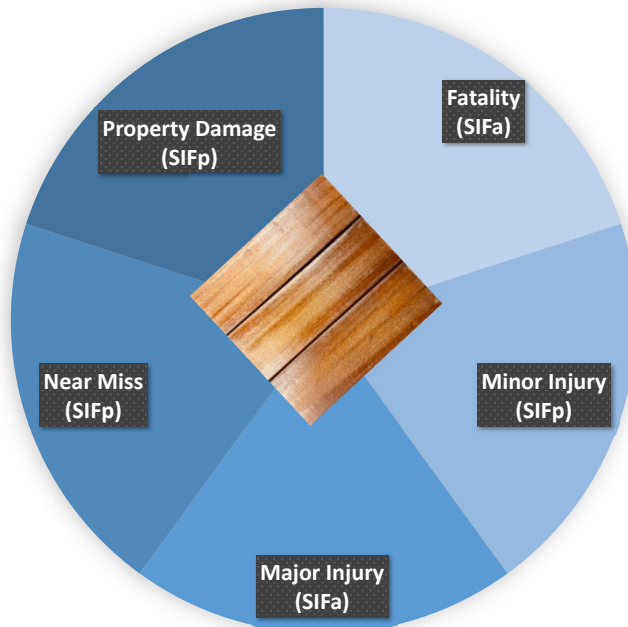


**“All models are wrong,
but some are useful”**

- George Box



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Precursors Converge

Can you think of a riskier situation?

- Employee's second day on the job
- New to the industry and company
- No training or orientation
- No supervisor or new supervisor
- Working alone
-And assigned to put on a harness and unload trash from the edge of the 12th floor of the building.
-OR properly install exterior glazing.

**Setup to
screw up!**



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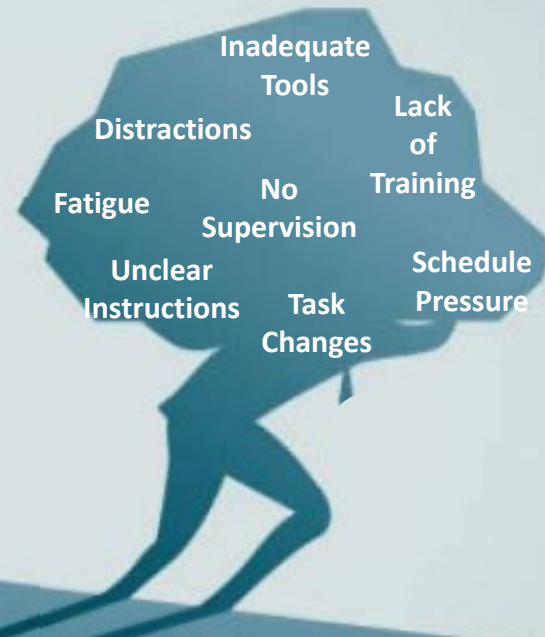
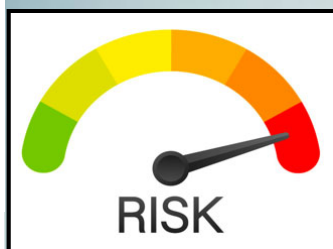
Precursors Converge



Ref: Axios



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Don't Skip the Deep Dive



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Deep Dives

Early

What precursors were present?

Intermediate

What evidence or explanations do we have to support those findings?

Advanced

Why did the precursor condition arise and survive?

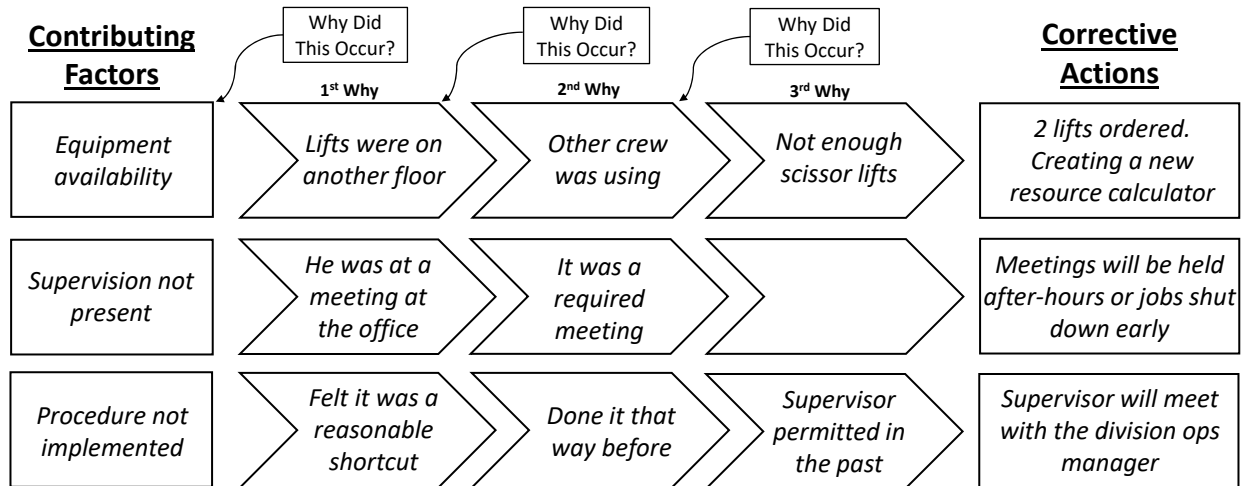


Ref: Saunders Construction



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Deep Dives



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Unrealistic budgets? Work out of market specialty? New estimator or project manager?

New key trade partners? Incomplete design/drawings? Limited space?

What are the precursors for a bad project / job?

New client or architect? Record volume? New methods or materials?

Workforce issues? New superintendent? Unrealistic schedule?

Largest project award? Work out of geographic market?



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Precursors Can Be Identified Pre-Incident



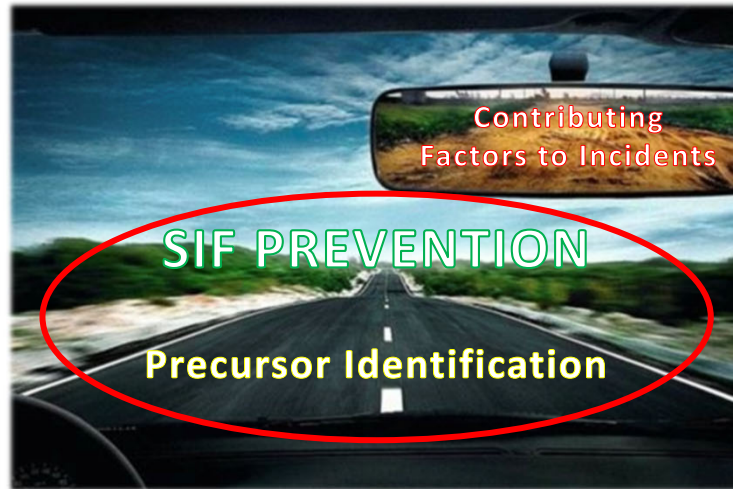
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**If we understand the ‘why’ behind
our incidents (precursors),
what should we do with this
knowledge?**



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Precursors Can Be Identified Pre-Incident



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Ref: Zion Realty

Precursors Can Be Identified Pre-Incident



Ref: Goodfellow Brothers

- Pre-Con & Pre-Task Planning
- Inspections
- Leadership Engagements
- Surveys
- Interviews
- Learning Teams
- Near Miss & Good Catch



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Pre-Task Planning



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Pre-Task Planning

DUMB

- What doesn't make sense to you?

DANGEROUS

- What feels risky about the work?

DIFFICULT

- What tasks are hard to do, or do well?

DIFFERENT

- What has changed or is changing?



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Ref. Learning Teams, Inc.

Pre-Task Planning

Task Steps	<u>Precursor</u> Hazards	Controls
	Fatigued employee	Switch their task, send home
	Infrequent task	Get procedure, call safety
	New/unplanned activity	Stop, crew re-huddles
	Lack of proper equipment	Order or go get it!
	New "green" crew members	Identify mentors/buddies
	Approaching critical deadlines	Add'l crew members or supers
	Employee missing training	Call safety, how did this happen?

(Identifying)

(Discussing, Avoiding or Managing)



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Inspections

vs.

Leadership Engagements



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Inspections

“Things seem to be going pretty good out on site”



How “things” are actually going



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Leadership Engagement

What is a Leadership Engagement?

A conversation between a leader and front-line employee, supervisor, or small group of employees.

The engagement can encompass all aspects of the work but should remain focused on potential risks.

Leadership Engagement

- What's on your mind?
- What can make this dangerous?
- Are there any tools or equipment that could make this easier?
- What are people concerned about, but no one is talking about?
- What's working/not working?



Ref: Rogers O'Brien

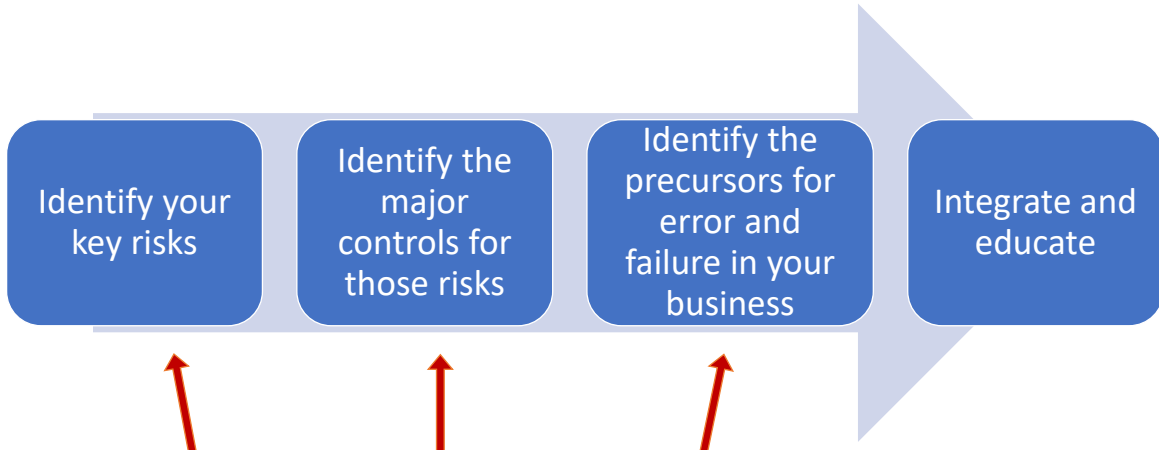


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Wrap-Up

Implementation



If you don't know what they are - ask your employees



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Variability is Normal

Perfection is a fantasy

We do not have (and will never have) :

- Perfect planning and scheduling
- Perfect communication
- Perfect procedures
- Perfect conditions
- Perfect tools and equipment
- Perfect people



Ref: Depositphotos



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